Year 9

REVISION PACK

Unit 2

Name: _____

Class: _____

Unit 2 · Prisms (Lessons 2.1, 2.2)

Worked example: triangular prisms

A triangular prism has a right-angled triangle with sides 3 cm, 4 cm and 5 cm. Its length is 10 cm.

(a) Work out the volume of the prism. (b) Work out the total surface area of the prism.

(a)	Step 1	Area of the triangle	$\frac{1}{2} \times 3 \times 4 = 6$
	Step 2	Area × length	$6 \times 10 = 60 \text{ cm}^3$
(b)	Step 1	Two triangles	$2 \times 6 = 12$
	Step 2	Three rectangles	$3 \times 10 + 4 \times 10 + 5 \times 10 = 120$
	Step 3	Two triangles + rectangles	$12 + 120 = 132 \text{ cm}^2$

Practice: triangular prisms

1.1

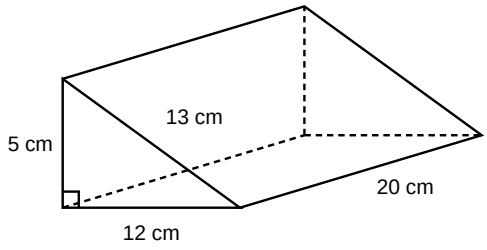


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

1.2

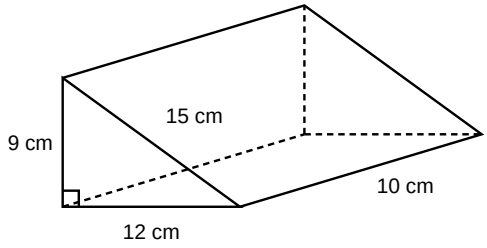


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

1.3

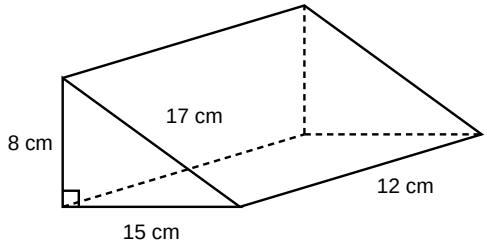


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

Practice: triangular prisms (continued)

1.4

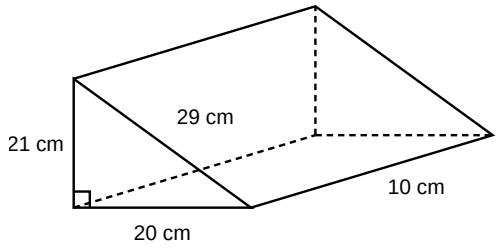


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

1.5

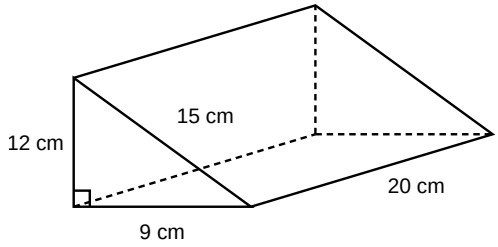


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

1.6

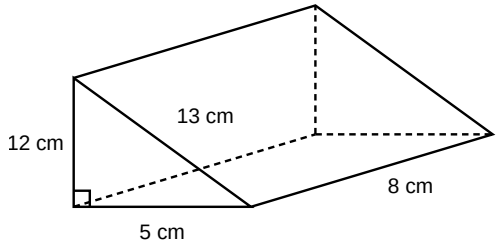


Diagram **NOT** accurately drawn

Here is a triangular prism.
(a) Work out the volume of the prism.
(b) Work out the total surface area of the prism.

Unit 2 · Prisms (Lessons 2.1, 2.2)

Worked example: litres in a tank

A tank is a cuboid 20 cm by 30 cm by 50 cm. 1 litre = 1000 cm³. How many litres of water does the tank hold when full?

Step 1	Volume	$20 \times 30 \times 50 = 30\,000\ \text{cm}^3$
Step 2	cm³ to litres: ÷ 1000	$30\,000 \div 1000 = 30$
Step 3	Answer	$30\ \text{litres}$

Practice: litres in a tank

1.7

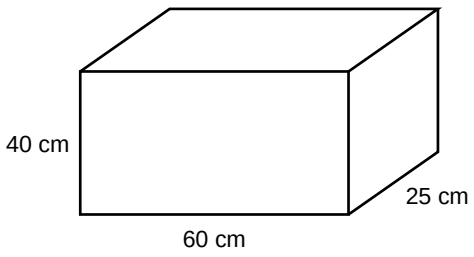


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

1.8

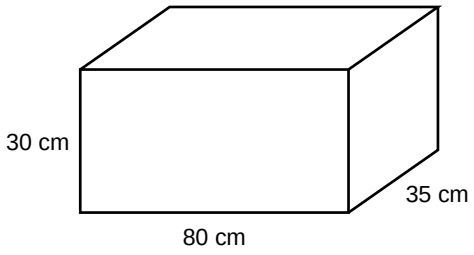


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

1.9

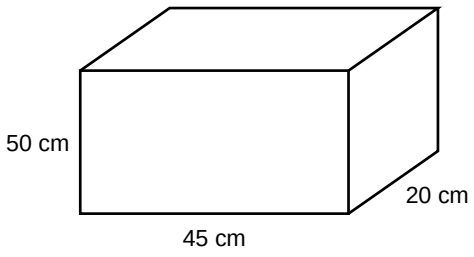


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

Practice: litres in a tank (continued)

1.10

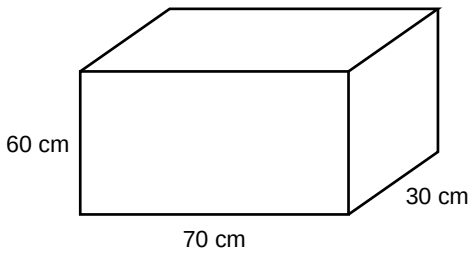


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

1.11

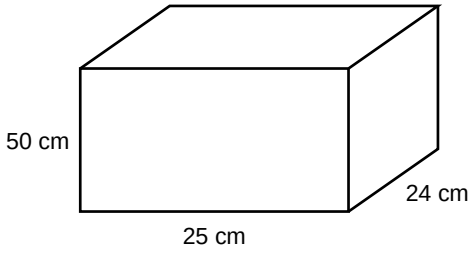


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

1.12

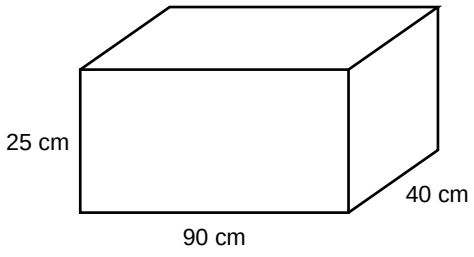


Diagram **NOT** accurately drawn

A tank is in the shape of a cuboid.
1 litre = 1000 cm³
How many litres of water does the tank hold when full?

Unit 2 · Circles (Lessons 2.3, 2.4)

Worked example: circumference

A circle has radius 6 cm. Work out the circumference of the circle. Give your answer correct to 1 decimal place.

Step 1	$C = 2\pi r$	$2 \times \pi \times 6 = 37.69911\ldots$
Step 2	Answer	$37.7\ \text{cm}$

Practice: circumference

2.1

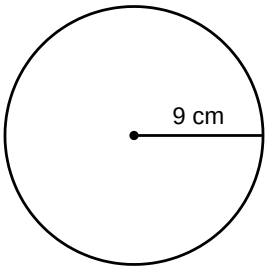


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

2.2

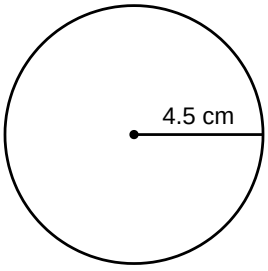


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

2.3

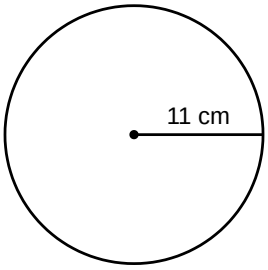


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

Practice: circumference (continued)

2.4

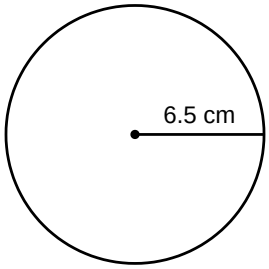


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

2.5

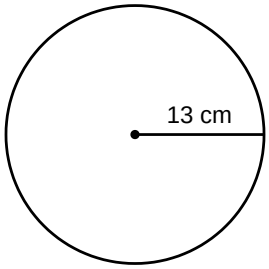


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

2.6

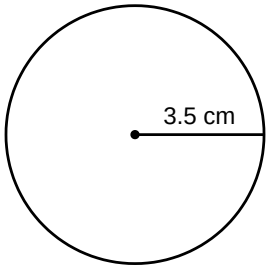


Diagram **NOT** accurately drawn

Work out the circumference of this circle.
Give your answer correct to 1 decimal place.

Unit 2 · Circles (Lessons 2.3, 2.4)

Worked example: area of a circle

A circle has diameter 8 cm. Work out the area of the circle. Give your answer correct to 1 decimal place.

Step 1	Radius = diameter ÷ 2	$8 \div 2 = 4$
Step 2	$A = \pi r^2$	$\pi \times 4^2 = 50.26548\ldots$
Step 3	Answer	$50.3\ \text{cm}^2$

Practice: area of a circle

2.7

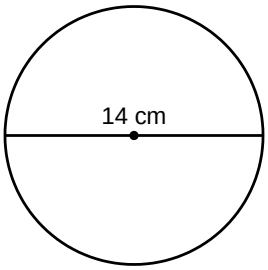


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

2.8

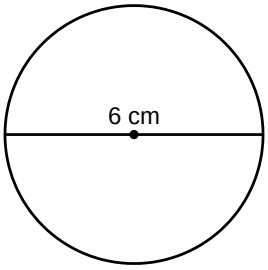


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

2.9

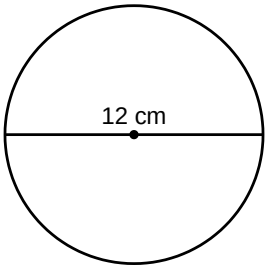


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

Practice: area of a circle (continued)

2.10

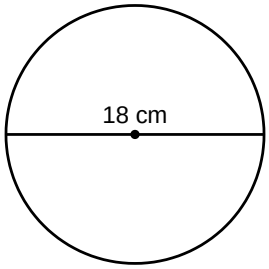


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

2.11

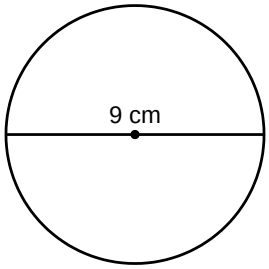


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

2.12

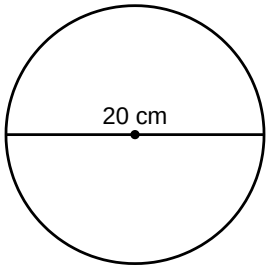


Diagram **NOT** accurately drawn

Work out the area of this circle.
Give your answer correct to 1 decimal place.

Unit 2 · Circles (Lessons 2.3, 2.4)

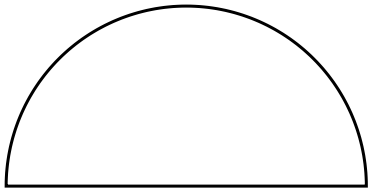
Worked example: perimeter of a semicircle

A semicircle has diameter 10 cm. Work out the perimeter of the semicircle. Give your answer correct to 1 decimal place.

Step 1	Half the circumference	$\pi \times 10 \div 2 = 15.70796\ldots$
Step 2	Add the diameter	$15.70796\ldots + 10 = 25.70796\ldots$
Step 3	Answer	$25.7\ \text{cm}$

Practice: perimeter of a semicircle

2.13

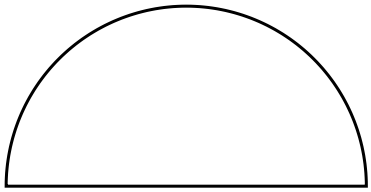


16 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

2.14

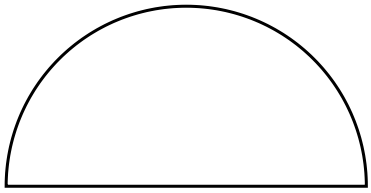


9 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

2.15



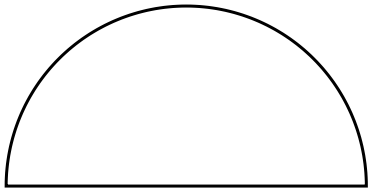
20 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

Practice: perimeter of a semicircle (continued)

2.16

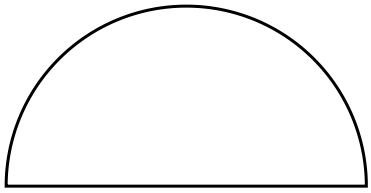


14 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

2.17

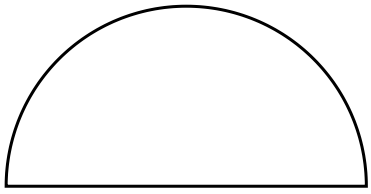


7 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

2.18



18 cm

Diagram **NOT** accurately drawn

The diagram shows a semicircle.
Work out the perimeter of the semicircle.
Give your answer correct to 1 decimal place.

Unit 2 · Circles (Lessons 2.3, 2.4)

Worked example: radius from the area

A circle has an area of 80 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

Step 1	$A = \pi r^2$	$\pi r^2 = 80$
Step 2	Divide by π	$r^2 = 80 \div \pi = 25.46479\ldots$
Step 3	Square root	$r = \sqrt{25.46479\ldots} = 5.046265\ldots$
Step 4	Answer	5.0 cm

Practice: radius from the area

2.19 A circle has an area of 40 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

2.20 A circle has an area of 120 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

2.21 A circle has an area of 65 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

2.22 A circle has an area of 30 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

2.23 A circle has an area of 200 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

2.24 A circle has an area of 90 cm². Work out the radius of the circle. Give your answer correct to 1 decimal place.

Worked example: volume and surface area of a cylinder

A solid cylinder has radius 3 cm and height 10 cm.

(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.

(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

(a)	Step 1	$\pi r^2 h$	$\pi \times 3^2 \times 10 = 282.7433\ldots$
	Step 2	Answer	$282.7\ \text{cm}^3$
(b)	Step 1	Two circles	$2 \times \pi \times 3^2 = 56.54867\ldots$
	Step 2	Curved surface	$2 \times \pi \times 3 \times 10 = 188.4956\ldots$
	Step 3	Ends + curved surface	$56.54867\ldots + 188.4956\ldots = 245.0442\ldots$
	Step 4	Answer	$245.0\ \text{cm}^2$

Practice: volume and surface area of a cylinder

3.1

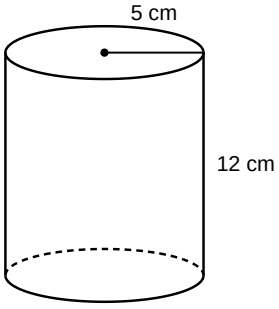


Diagram **NOT** accurately drawn

A solid cylinder has radius 5 cm and height 12 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

3.2

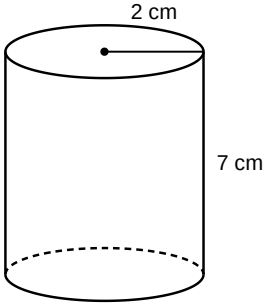


Diagram **NOT** accurately drawn

A solid cylinder has radius 2 cm and height 7 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

3.3

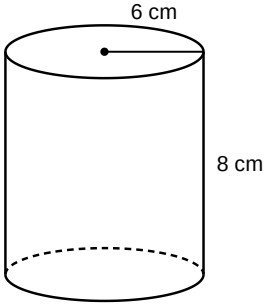


Diagram **NOT** accurately drawn

A solid cylinder has radius 6 cm and height 8 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

Practice: volume and surface area of a cylinder (continued)

3.4

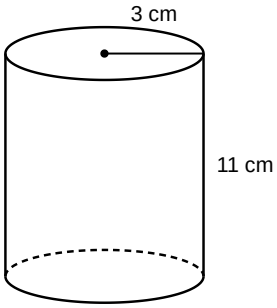


Diagram **NOT** accurately drawn

A solid cylinder has radius 3 cm and height 11 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

3.5

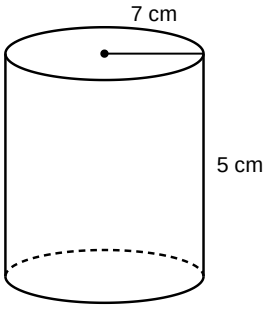


Diagram **NOT** accurately drawn

A solid cylinder has radius 7 cm and height 5 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

3.6

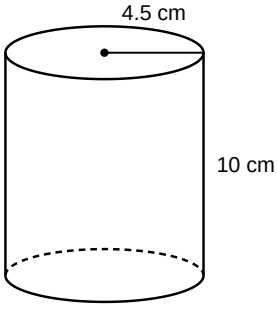


Diagram **NOT** accurately drawn

A solid cylinder has radius 4.5 cm and height 10 cm.
(a) Work out the volume of the cylinder. Give your answer correct to 1 decimal place.
(b) Work out the total surface area of the cylinder. Give your answer correct to 1 decimal place.

Worked example: finding the hypotenuse

The two shorter sides of a right-angled triangle are 9 cm and 12 cm. Work out the length of the hypotenuse, x.

Step 1	Square and add	$x^2 = 9^2 + 12^2 = 225$
Step 2	Square root	$x = \sqrt{225}$
Step 3	Answer	$x = 15$ cm

Practice: finding the hypotenuse

4.1

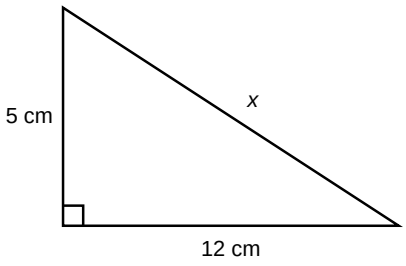


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.

4.2

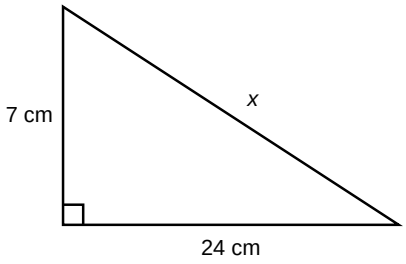


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.

4.3

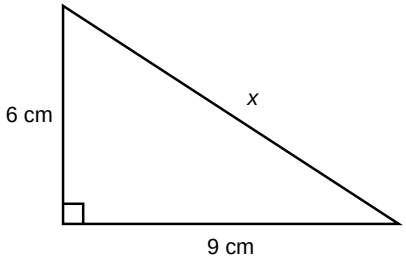


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.
Give your answer correct to 1 decimal place.

Practice: finding the hypotenuse (continued)

4.4

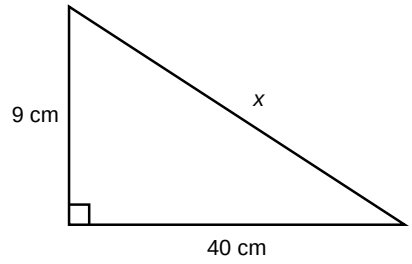


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.

4.5

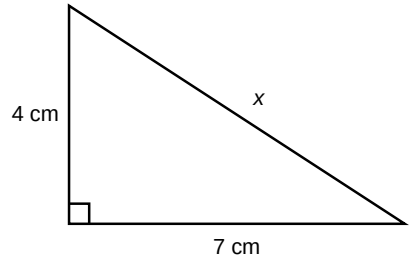


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.
Give your answer correct to 1 decimal place.

4.6

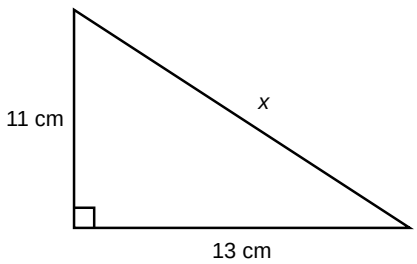


Diagram **NOT** accurately drawn

Work out the length of the hypotenuse.
Give your answer correct to 1 decimal place.

Worked example: finding a shorter side

The hypotenuse of a right-angled triangle is 25 cm. One shorter side is 24 cm. Work out the length of the other side, y.

Step 1	Square and subtract	$y^2 = 25^2 - 24^2 = 49$
Step 2	Square root	$y = \sqrt{49}$
Step 3	Answer	$y = 7$ cm

Practice: finding a shorter side

4.7

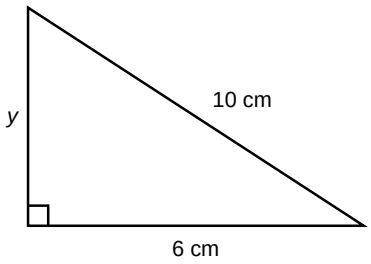


Diagram **NOT** accurately drawn

Work out the value of y.

4.8

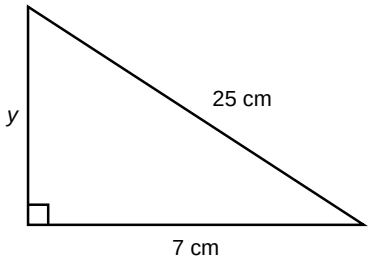


Diagram **NOT** accurately drawn

Work out the value of y.

4.9

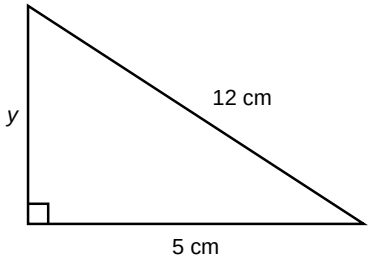


Diagram **NOT** accurately drawn

Work out the value of y.

Give your answer correct to 1 decimal place.

Practice: finding a shorter side (continued)

4.10

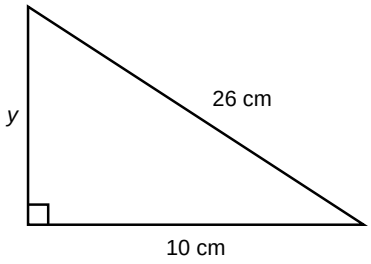


Diagram **NOT** accurately drawn

Work out the value of y.

4.11

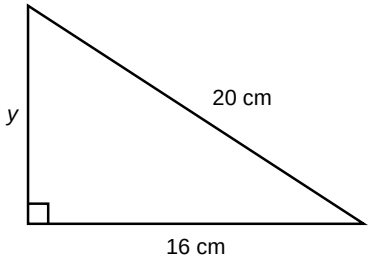


Diagram **NOT** accurately drawn

Work out the value of y.

4.12

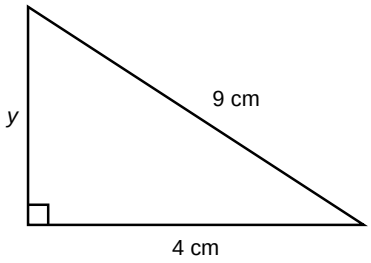


Diagram **NOT** accurately drawn

Work out the value of y.

Give your answer correct to 1 decimal place.

Worked example: the diagonal of a rectangle

A rectangle is 6 cm long and 3 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

Step 1	Square and add	$d^2 = 6^2 + 3^2 = 45$
Step 2	Square root	$d = \sqrt{45} = 6.708204\ldots$
Step 3	Answer	6.7 cm

Practice: the diagonal of a rectangle

4.13 A rectangle is 8 cm long and 5 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

4.14 A rectangle is 12 cm long and 7 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

4.15 A rectangle is 10 cm long and 3 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

4.16 A rectangle is 7 cm long and 2 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

4.17 A rectangle is 11 cm long and 6 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

4.18 A rectangle is 15 cm long and 4 cm wide. Work out the length of a diagonal of the rectangle. Give your answer correct to 1 decimal place.

Unit 2 · Units of volume (Lesson 2.2)

Worked example: litres into cm³

Convert 2.5 litres into cm³.

Step 1	1 litre = 1000 cm³	$2.5 \times 1000 = 2500$
Step 2	Answer	$2500\ \text{cm}^3$

Practice: litres into cm³

5.1 Convert 4.2 litres into cm³.

5.2 Convert 0.75 litres into cm³.

5.3 Convert 6 litres into cm³.

5.4 Convert 1.35 litres into cm³.

5.5 Convert 0.4 litres into cm³.

5.6 Convert 12.5 litres into cm³.

Unit 2 · Distance between two points (Lesson 2.6)

Worked example: the distance between two points

Find the distance between the points $A(1, 2)$ and $B(7, 10)$. You could use a sketch to help.

Step 1	Horizontal distance	$7 - 1 = 6$
Step 2	Vertical distance	$10 - 2 = 8$
Step 3	Use Pythagoras' theorem	$\sqrt{6^2 + 8^2} = \sqrt{100}$
Step 4	Answer	10 units

Practice: the distance between two points

7.1 Find the distance between the points $P(2, 1)$ and $Q(5, 5)$.
Give your answer correct to 3 significant figures where necessary.

7.2 Find the distance between the points $A(-3, 1)$ and $B(5, 7)$.
Give your answer correct to 3 significant figures where necessary.

7.3 Find the distance between the points $C(0, -2)$ and $D(5, 10)$.
Give your answer correct to 3 significant figures where necessary.

7.4 Find the distance between the points $E(-4, -1)$ and $F(2, 3)$.
Give your answer correct to 3 significant figures where necessary.

7.5 Find the distance between the points $G(1, 6)$ and $H(-2, -1)$.
Give your answer correct to 3 significant figures where necessary.

7.6 Find the distance between the points $J(-5, 2)$ and $K(3, -4)$.
Give your answer correct to 3 significant figures where necessary.